PREDICTIVE ANALYTICS FOR CUSTOMER BEHAVIOR

Customer Churn Prediction using Machine Learning

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PROJECT OVERVIEW

1.1 Introduction

Customer retention is one of the major challenges faced by service-based industries. Acquiring a new customer costs significantly more than retaining an existing one. Therefore, organizations need intelligent systems capable of predicting customer churn before it occurs.

This project applies data analytics and machine learning techniques to analyze customer behaviour using the IBM Telco Customer Churn dataset. The study is divided into two phases. The first phase focuses on data cleaning, preprocessing, exploratory data analysis (EDA), and extracting meaningful insights from customer data. The second phase develops predictive machine learning models to forecast customer churn and recommends business strategies to improve customer retention.

The project demonstrates how customer demographic information, service subscriptions, contract details, payment methods, and billing history can be utilized to identify customers at risk of leaving the company.

CHAPTER 2

PROBLEM STATEMENT

Customer churn directly impacts company revenue and long-term profitability. Businesses often lose valuable customers because they fail to recognize early warning signs.

The objective of this project is to build a predictive analytics solution capable of identifying customers who are likely to churn so that organizations can take proactive retention measures.

CHAPTER 3

OBJECTIVES

The primary objectives of the project are:

- Understand customer behaviour using exploratory data analysis.
- Clean and preprocess customer data.
- Identify important factors influencing churn.
- Perform feature engineering.
- Build multiple machine learning models.
- Compare model performance.
- Select the best predictive model.
- Provide data-driven business recommendations.

CHAPTER 4

DATASET DESCRIPTION

1. Dataset Used

   IBM Telco Customer Churn Dataset

2. Number of Records

   7043 unique customers

3. Original Features

   33

4. Features Include

   - Customer demographics
   - Service subscriptions
   - Contract information
   - Billing details
   - Customer Lifetime Value
   - Churn information

5. Target Variable - **Churn Value**

CHAPTER 5

TOOLS & TECHNOLOGIES

Software	Purpose
Python	Programming Language
Jupyter Notebook	Development Environment
Pandas	Data Manipulation
NumPy	Numerical Computing
Matplotlib	Visualization
Seaborn	Statistical Visualization
Scikit-Learn	Machine Learning
VS Code	Development
GitHub	Version Control

CHAPTER 6

PROJECT METHODOLOGY

Flowchart

Dataset

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Data Collection

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Data Cleaning

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EDA

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Model Building

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Business Recommendations

6.1 Data Collection

The first stage of the project involved collecting the Telco Customer Churn dataset. The dataset contains customer demographic information, subscribed services, billing details, account information, and churn status. It consists of 7,043 customer records with 33 attributes, making it suitable for customer behavior analysis and churn prediction. The dataset includes numerical, categorical, and geographical variables, providing a comprehensive view of customer characteristics.

Activities Performed

- Obtained the Telco Customer Churn dataset.
- Loaded the dataset into Python using the Pandas library.
- Inspected the dataset structure.
- Examined the number of rows and columns.
- Identified feature names and data types.
- Verified the target variable (Churn Label).

Purpose

The objective of this stage was to gather reliable customer information for further analysis and predictive modeling.

6.2 Data Cleaning and Preprocessing

Raw datasets often contain missing values, duplicate records, inconsistent formats, and unnecessary attributes that can negatively impact model performance. Therefore, data preprocessing was performed to improve data quality before analysis.

The following preprocessing tasks were carried out:

➢ Missing Value Treatment

The Total Charges column contained blank values for newly joined customers with zero tenure. These values were converted into numeric format and replaced with appropriate values to maintain consistency.

➢ Handling Null Values

The Churn Reason column contained missing values for active customers. Since these customers had not left the company, the missing values were replaced with "Not Applicable" instead of leaving them blank.

➢ Duplicate Record Checking

The Customer ID column was checked for duplicate records. No duplicate customer records were found in the dataset.

➢ Removing Unnecessary Features

Redundant columns such as Count and Lat Long were removed because they provided no additional analytical value.

➢ Data Type Conversion

Several columns were converted into appropriate numerical and categorical formats to facilitate statistical analysis and machine learning.

Purpose

The purpose of preprocessing was to ensure that the dataset was clean, consistent, and suitable for exploratory analysis and predictive modeling.

6.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand customer behavior, identify patterns, detect trends, and discover relationships between different variables. Various statistical summaries and visualizations were created using Matplotlib and Seaborn.

The analysis included:

- Customer churn distribution
- Customer tenure analysis
- Monthly charges distribution
- Contract type analysis
- Internet service analysis
- Payment method analysis
- Customer Lifetime Value (CLTV)
- Correlation analysis
- Demographic analysis
- Add-on services analysis
- Customer segmentation

Visualizations such as bar charts, count plots, histograms, box plots, and heatmaps were generated to understand customer behavior. The EDA revealed important churn drivers including contract type, tenure, monthly charges, payment methods, and service subscriptions.

Purpose

The main objective of EDA was to identify hidden patterns and determine which customer attributes have the greatest influence on churn.

6.4 Feature Engineering

Feature engineering improves the predictive capability of machine learning models by creating meaningful input variables.

The following activities were performed:

• Converted categorical variables into numerical values using encoding techniques.

• Created customer value segments based on Customer Lifetime Value (CLTV).

• Selected the most relevant features affecting customer churn.

• Removed irrelevant attributes that did not contribute to prediction.

• Prepared the feature matrix (X) and target variable (y).

Purpose

Feature engineering enhances model accuracy by providing informative and relevant features while reducing unnecessary information.

6.5 Model Building

After preprocessing and feature engineering, machine learning models were developed to predict customer churn.

The dataset was divided into training and testing subsets using an 80:20 ratio. Several classification algorithms were considered, including:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

The models were trained using historical customer data to classify whether a customer is likely to churn or remain with the company.

Purpose

The objective of model building was to create an accurate predictive model capable of identifying customers who are at risk of leaving.

6.6 Model Evaluation

The performance of the trained models was evaluated using standard classification metrics. The evaluation included:

• Accuracy

• Precision

• Recall

• F1-Score

• ROC-AUC Score

• Confusion Matrix

These metrics were used to compare different machine learning models and identify the best performing model for churn prediction.

Purpose

The goal of model evaluation was to measure prediction performance and ensure that the selected model generalizes well to unseen customer data.

6.7 Business Recommendations

Based on the exploratory analysis and predictive modeling results, several business recommendations were proposed to reduce customer churn and improve customer retention. Key recommendations include:

• Focus retention efforts on new customers during their first two years.

• Encourage customers to switch from month-to-month contracts to long-term plans.

• Improve the quality of customer support services.

• Promote automatic payment methods to reduce churn.

• Bundle value-added services such as online security and technical support.

• Prioritize retention campaigns for high-value customers with high churn risk.

• Improve customer satisfaction by addressing service quality and pricing concerns.

Purpose

The recommendations help organizations make data-driven decisions that improve customer satisfaction, increase loyalty, and reduce revenue loss caused by customer churn.

CHAPTER 7

PHASE 1 – CUSTOMER BEHAVIOR ANALYSIS & EDA

7.1 Dataset Exploration

The following data quality issues were identified and resolved:

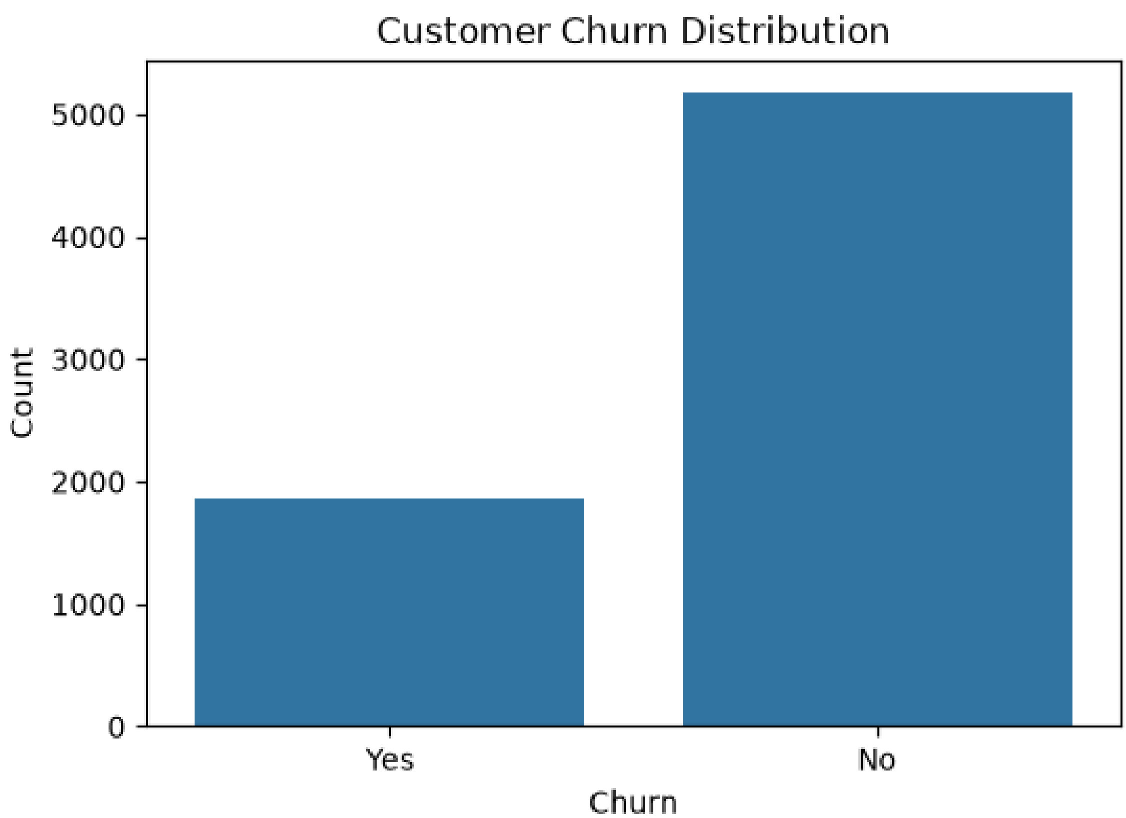
• 'Total Charges' contained 11 blank entries for customers with zero tenure these were converted to numeric and filled with 0 or the equivalent monthly charge.

• 'Churn Reason' was null for all active (non-churned) customers by design; these were explicitly labeled 'Not Applicable' rather than left as missing.

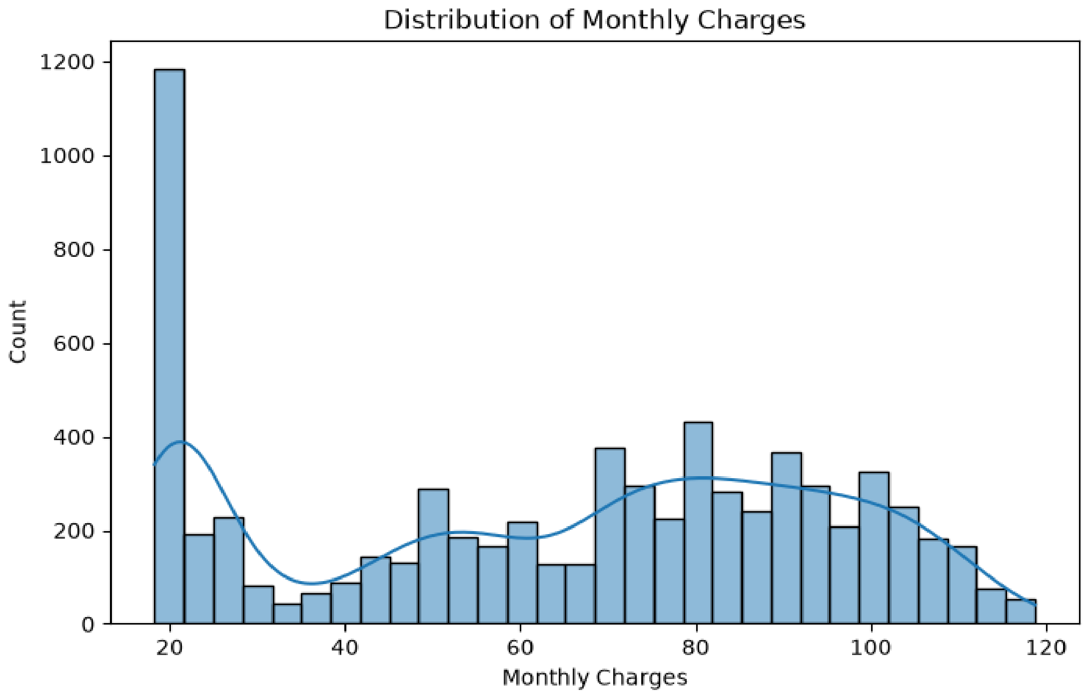
• Redundant columns were dropped since 'Latitude' and already exist as numeric. No duplicate customer records were found in the dataset.

7.2 Exploratory Data Analysis

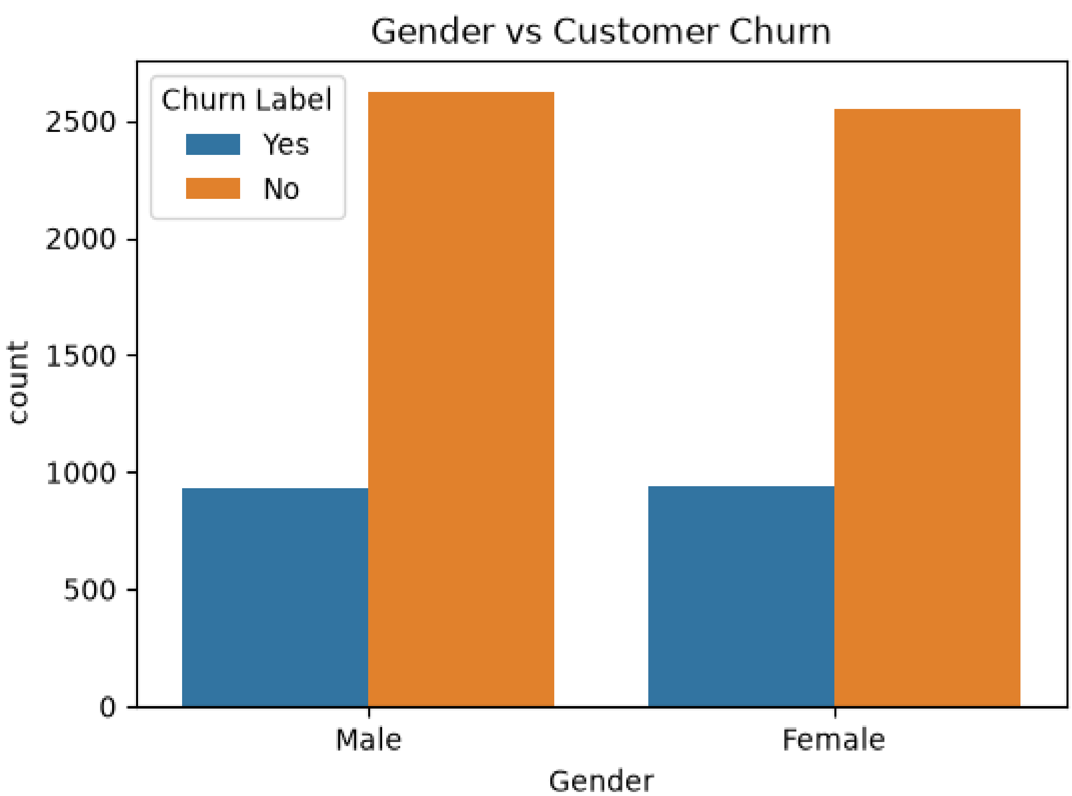
1) Customer Churn Distribution



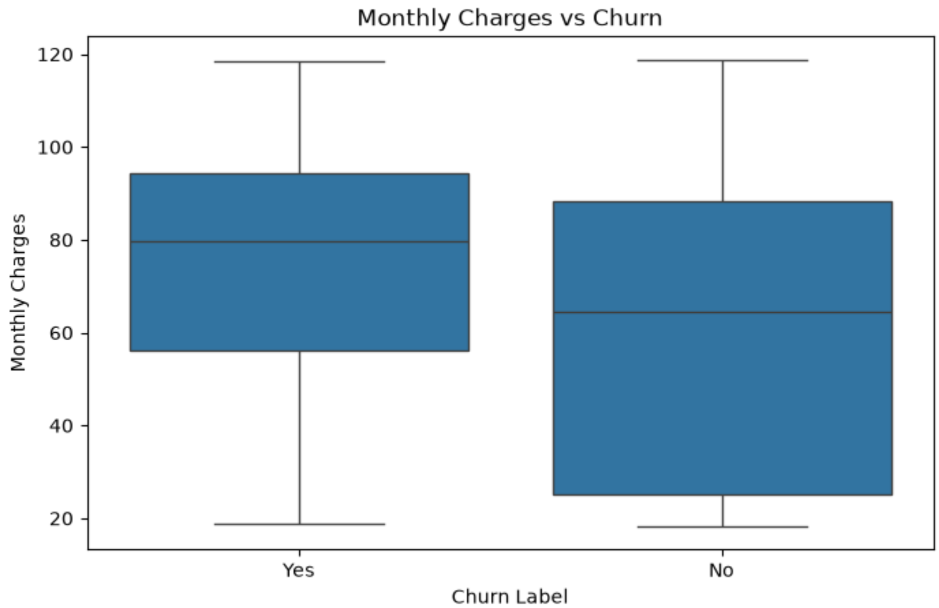
2) Monthly Charges



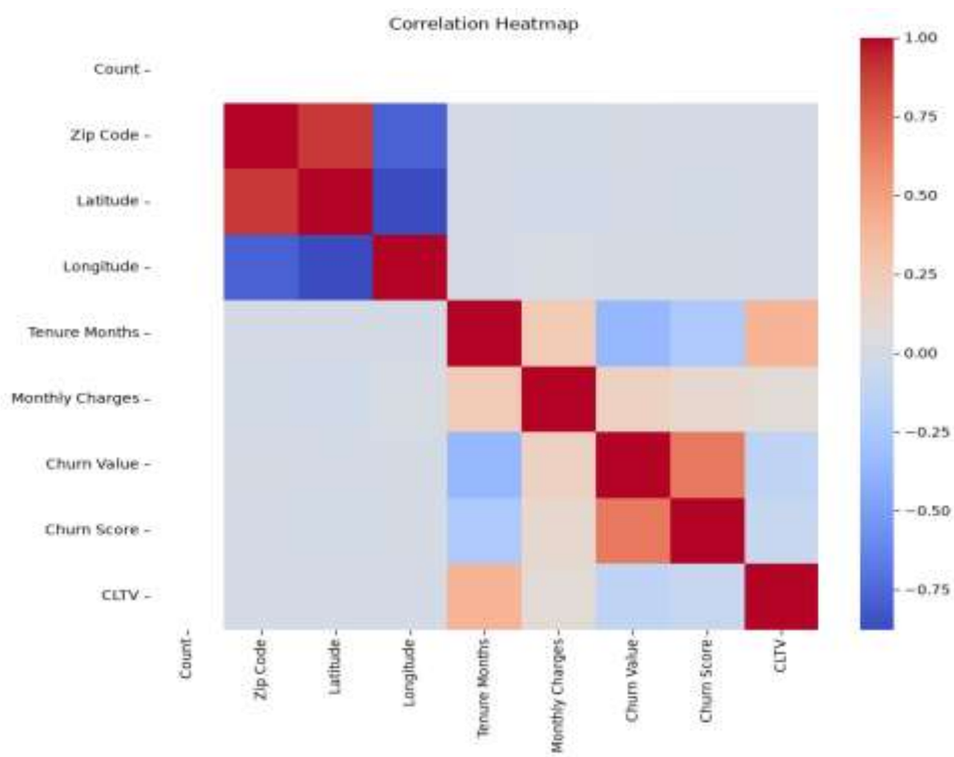
3) Gender vs Churn



4) Monthly Charges Analysis



5) Correlation Analysis



7.3 Key Findings

- Overall churn rate is approximately 26.5%.
- Customers with month-to-month contracts exhibit the highest churn.
- Fiber optic users churn more frequently than DSL users.
- Electronic check customers have a higher churn rate.
- Longer tenure customers are less likely to churn.
- Higher CLTV customers are more loyal.

PHASE 2 – PREDICTIVE MODELING

Objective

Phase 1 of this project focused on cleaning the IBM Telco Customer Churn dataset and performing exploratory data analysis. Phase 2 builds on this foundation by engineering predictive features, training multiple classification models to predict customer churn, evaluating and comparing their performance, and translating the model results into actionable business recommendations for customer retention.

Chapter 8

8.1 Feature Engineering

Feature engineering is the process of transforming raw data into meaningful features that improve the performance of machine learning models. Well-designed features help the models recognize hidden patterns in customer behaviour and produce more accurate predictions.

In this project, feature engineering was performed after completing data cleaning and exploratory data analysis. The objective was to prepare the dataset in a format suitable for machine learning algorithms while preserving the important information related to customer churn.

8.1.1Feature Engineering Steps

The following feature engineering techniques were applied:

- Removed unnecessary columns such as **Customer ID**, **Churn Label**, and **Churn Reason**, since they either uniquely identify customers or leak information about the target variable.
- Converted **Total Charges** into a numeric data type for mathematical analysis.
- Selected relevant customer attributes such as tenure, monthly charges, contract type, payment method, internet service, and customer lifetime value (CLTV).
- Separated the dataset into:
  - **Feature Matrix (X)** – Independent variables used for prediction.
  - **Target Variable (y)** – Churn Value, where:
    - **0 = Customer Retained**
    - **1 = Customer Churned**

8.1.2 Encoding and Scaling

Machine learning algorithms require numerical input. Since the dataset contains several categorical variables such as gender, contract type, internet service, and payment method, these variables were converted into numerical values using Label

Original Value	**Encoded Value**
Yes	**1**
No	**0**
Male	**1**
Female	**0**

Train-Test Split

Dataset	**Shape**
Training Data	**5625 × 28**
Testing Data	**1407 × 28**

- **80%** of the data was used for training.
- **20%** of the data was used for testing.

8.2 Machine Learning Models

8.2.1 Decision Tree Classifier

A Decision Tree is a supervised machine learning algorithm used for classification and regression tasks. It predicts the target variable by splitting the dataset into smaller subsets based on feature values. The model constructs a tree-like structure consisting of decision nodes and leaf nodes.

8.2.2 Logistic Regression

Logistic Regression is a supervised classification algorithm that predicts the probability of a customer belonging to a particular class. It is widely used for binary classification problems such as customer churn prediction.

8.2.3 Random Forest Classifier

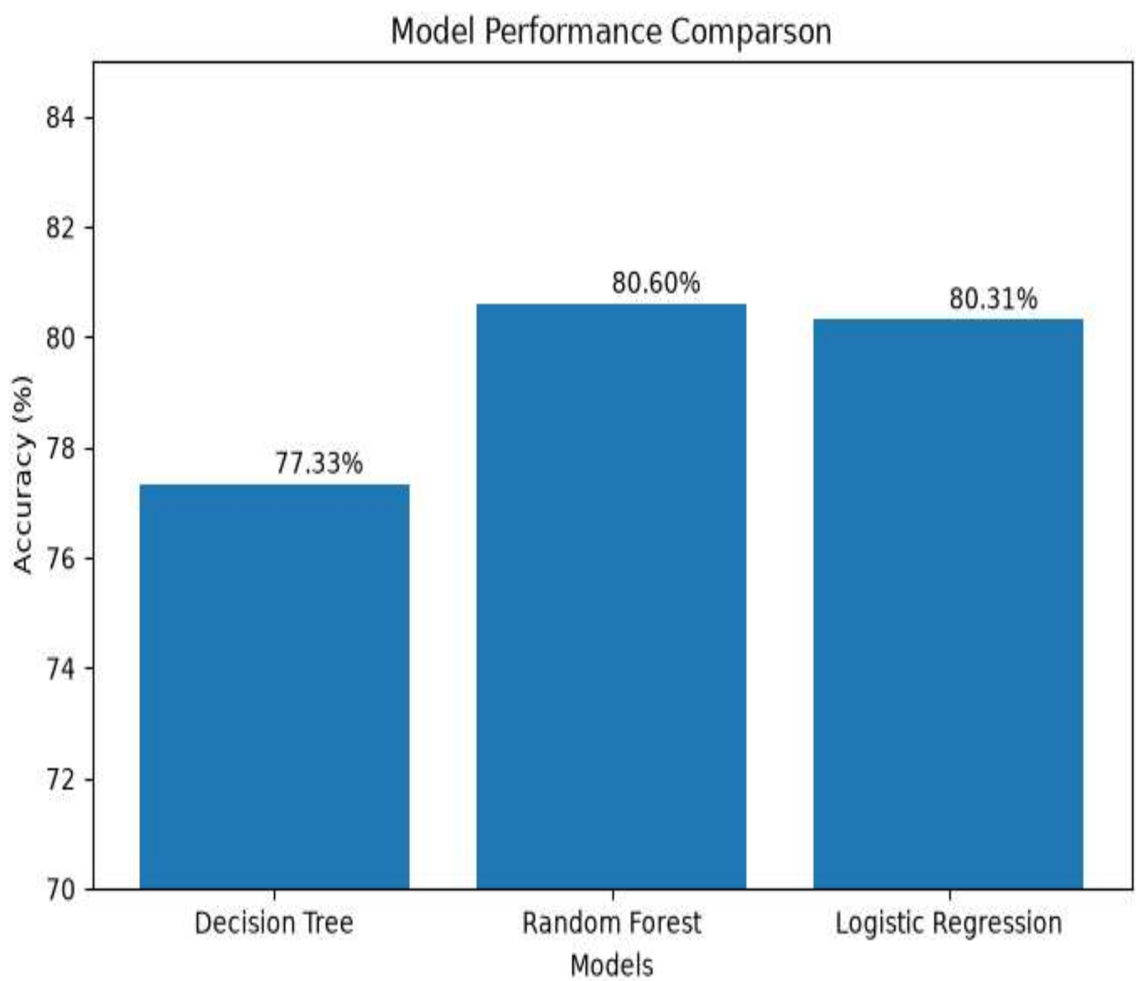
Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to improve prediction accuracy and reduce overfitting. Each tree is trained on a random subset of the dataset, and the final prediction is obtained using majority voting.

8.3 Model Evaluation

Machine Learning Model	Accuracy	Precision	Recall	F1-Score	Confusion Matrix(TN, FP, FN, TP)	Evaluation
Decision Tree	**77.33%**	0.57	0.64	0.60	[[849, 184], [135, 239]]	Provides good interpretability but lower prediction accuracy compared to Random Forest.
Random Forest	**80.60%**	0.64	0.49	0.56	[[928, 105], [190, 184]]	Achieved the highest accuracy with better generalization, making it the best-performing model.
Logistic Regression	**80.31%**	0.63	**0.61**	**0.62**	[[849 184], [135 239]]	Performs well as a baseline model with good interpretability and competitive accuracy.

8.4 Model Performance Comparison

Model	Accuracy
Decision Tree	77.33%
Logistic Regression	80.31%
Random Forest	80.60%



8.5 Feature Importance

Feature	Importance Score	Interpretation
Tenure Months	0.1003	Customers with longer tenure are less likely to churn.
Total Charges	0.0945	Total spending reflects customer commitment and loyalty.
Monthly Charges	0.0882	Higher monthly charges increase the likelihood of churn.
Contract Type	0.0728	Customers with long-term contracts are more likely to remain.
CLTV	0.0669	Customers with higher Customer Lifetime Value tend to stay longer.
Zip Code	0.0640	Geographic location has a moderate influence on customer behaviour.
City	0.0625	Customer location contributes slightly to churn prediction.
Longitude	0.0617	Geographic features have a minor impact on churn.
Lat Long	0.0588	Combined location information provides limited predictive value.

CHAPTER 9

BUSINESS RECOMMENDATIONS

Customer Retention

- Target month-to-month customers.
- Encourage long-term contracts.
- Improve customer support.

Personalized Marketing

- Special offers for new customers.
- Loyalty rewards for high-value customers.
- Personalized service recommendations.

Customer Engagement

- Improve service quality.
- Increase customer interaction.
- Monitor high-risk customers monthly.

CHAPTER 10

CONCLUSION

This project successfully analyzed customer behaviour using exploratory data analysis and machine learning techniques. Data preprocessing and feature engineering improved the quality of the dataset and enabled effective predictive modelling. Three classification algorithms—Decision Tree, Logistic Regression, and Random Forest—were evaluated for customer churn prediction.

Among them, **Random Forest** achieved the highest accuracy of **80.60%**, followed closely by **Logistic Regression** at **80.31%**, making Random Forest the most suitable model for predicting customer churn. The analysis identified key factors such as tenure, monthly charges, contract type, internet service, and customer lifetime value as significant contributors to churn.

The insights obtained from this project can help businesses design effective retention strategies, improve customer satisfaction, and make informed business decisions using predictive analytics.